



UiT

**THE ARCTIC
UNIVERSITY
OF NORWAY**

Home Exam

Home Exam in FYS-3033 - Deep Learning

Hand-out: Monday March 31, 2025, 09:00

Hand-in: Monday April 28, 2025, 14:00

The Home Exam contains 8 pages including this cover page

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Before You Start

Portfolio instructions

Your code should be submitted together with your report (see instructions below). **Further, please include a discussion of the results obtained and a discussion of the implementation in your report, which show that you understand what you are doing.**

The code should be commented in such a way that any person with programming knowledge should be able to understand how the program works. Like your report, the code must be your own individual work.

You are permitted to use deep learning frameworks such as Pytorch and Tensorflow. As there is a lot of code available online, please make sure that your report and code clearly show that you understand what you are doing.

Hand-in format

Please submit your report (in pdf format) to WISEflow and attach *one* single .zip file that contains two folders, one called doc that contains your report, and another one called src containing the code. The file name of the .zip file should follow the format homeexam_candidateXX.zip (replace XX with your candidate number obtained from WISEflow) for anonymity.

Please include your candidate number and the course name on the frontpage of your report.

Follow the hand-in instruction in Wiseflow. Upload the pdf as the main file and then attach your zip as an attachment. Note, the reports will be processed by a plagiarism checker and the **pdf file size must not exceed 15MB**.

Problem 1

In this problem, you will derive some of the theoretical results that are found in unsupervised deep learning.

- (1a) The goal of generative models is to sample from the underlying distribution, p_{data} , of data $\mathbf{x}$. Generative Adversarial Networks (GANs) achieve this goal by constructing a generator G , which tries to fool a discriminator D . The generator transforms a sample $\mathbf{z}$ from a prior noise distribution, $p_{\mathbf{z}}$, into a new distribution p_g . Given the loss that the discriminator is tasked to maximize

$$V(G, D) = \int_{\mathbf{x}} p_{\text{data}(\mathbf{x})} \log(D(\mathbf{x})) d\mathbf{x} + \int_{\mathbf{z}} p_{\mathbf{z}}(\mathbf{z}) \log(1 - D(g(\mathbf{z}))) d\mathbf{z}, \quad (1)$$

and assuming that the optimal solution has been reached ($p_g = p_{\text{data}}$). Show that the optimal solution for the discriminator given a fixed generator g is

$$D(\mathbf{x}) = \frac{1}{2}. \quad (2)$$

Finally, illustrate that this optimum corresponds to a value of $-\log(4)$.

- (1b) The reparameterization trick is a key component of Variational Autoencoders (VAEs). What is its purpose in VAEs? For VAEs the latent variable is sampled from a normal distribution with parameters $\mu_{z|x}$ and $\sigma_{z|x}$, where the parameters are learned by the encoder of the VAE. For the reparameterization trick, we perform our sampling operation as

$$z = \mu_{z|x} + \sigma_{z|x}\epsilon, \quad (3)$$

where $\epsilon \sim \mathcal{N}(0, 1)$. Show that z follows a normal distribution with parameters $\mu_{z|x}$ and $\sigma_{z|x}$.

- (1c) Show that the KL divergence between two univariate Bernoulli distributions $p(x) = \text{Bern}(p_1)$ and $q(x) = \text{Bern}(p_2)$ can be expressed as

$$KL(p, q) = \log \frac{(1 - p_1)}{(1 - p_2)} + p_1 \log \frac{p_1 (1 - p_2)}{p_2 (1 - p_1)}. \quad (4)$$

Discuss how this result can be used in VAEs.

Problem 2

You are working for the company Planes, Ships, and Trucks Inc., and have been tasked with developing an image classifier to automatically classify images of planes, ships and trucks. Your co-worker Elisabeth Wetzer has worked day and night to gather and label images that your company is interested in, and has provided you with a dataset (`Canvas/Files/homeexam/problem2.zip` (`training_data.npz`, `evaluation_data.npz`)) with the following content:

- 1500 labeled color images of size 96x96 for training.
- 3 classes; plane, ship, and truck.
- 2400 labeled color images of size 96x96 for validation.

Your job is to develop an algorithm that can successfully tackle the classification task and undoubtedly quadruple the revenue stream of Planes, Ships, and Trucks Inc.

Note: The data is stored in Numpy arrays (.npz extension), and can be loaded like shown below. Remember to replace the path your local directory.

```
import numpy as np
training_data = np.load('your/path/training_data.npz')
training_images = training_data['a']
training_labels = training_data['b']
```

- (2a) Based on the task and characteristics of the dataset, you decide that a VGG-11 with Batch Normalization after the convolutional layers should be a good start. Explain why. Note: Do not use pre-trained weights/finetuning in this exercise, nor the premade model from e.g. Torchvision. For details about the VGG-11 architecture, see column A of Table 1 in Simonyan and Zisserman (1).
- (2b) Explain Dropout (2).
- (2c) Include Dropout in the network from problem (2a) with a dropout rate of 0.5 after each of the two layers that contain 4096 filters, discuss results.

Your co-worker Harald Lykke Joakimsen is a hobby photographer and wants to help out with increasing the size of the database to improve the evaluation of the algorithm. Therefore, he has gone out and gathered new images (Canvas/Files/homeexam/problem2.zip (new_evaluation_data_with_labels.npz, new_evaluation_data_without_labels.npz')). However, in his eagerness to help, he also took images of birds, which is not interesting for your company (Planes, Ships, and Trucks Inc.). He managed to label some of the images, but a large portion of the dataset remains unlabeled. The content of new dataset is:

- 1600 labeled color images of size 96x96, containing both the known classes and a new unwanted class.
- 3 known classes; plane, ship, and truck, and one unwanted class; bird.
- 1600 unlabeled color images of size 96x96 containing both the known classes and the unwanted class. To use the new dataset for evaluation in the future, you must remove as many images of new the class as possible, while keeping as much as possible of the known classes. You get an idea that since your trained model has never seen this new class, it should be more uncertain about those samples. Therefore, you will investigate if uncertainty modeling can be used to filter out the unwanted images in the new dataset.

Note: The new data is stored as before, but the unlabeled dataset only has images and no labels.

```
import numpy as np
new_evaluation_data = np.load(
    'your/path/new_evaluation_data_with_labels.npz')

new_evaluation_images = new_evaluation_data['a'],
new_evaluation_labels = new_evaluation_data['b']

unlabeled_evaluation_data = np.load(
    'your/path/new_evaluation_data_without_labels.npz')

unlabeled_evaluation_images = unlabeled_evaluation_data['a'] # no 'b'.
```

(2d) Leverage the Dropout-mechanism in your trained network to create uncertainty estimates for each of the labeled images in the new dataset. You will use the the concept of entropy from information theory to indicate uncertainty as follows:

- For a single sample, do 10 forward passes with Dropout applied and calculate the mean softmax output across the 10 predictions.
- Calculate the entropy of the mean softmax output $\hat{\mathbf{y}}$ as:

$$H(\hat{\mathbf{y}}) = - \sum_{i=1}^{N_c} \hat{y}_i \log \hat{y}_i,$$

where N_c is the number of classes in your trained model.

Explain why the entropy of the softmax output can be interpreted as a notion of uncertainty. Make a histogram where you plot the entropy of the known classes as one distribution and the entropy of the new class as a second distribution. Calculate the mean of each distribution and mark it in the histogram. What do you observe?

Disclaimer: Uncertainty modeling is a challenging topic in deep learning. It is unlikely that you will be

able to find all images of the new class, and some images from the known classes will also most likely be removed.

- (2e) Using the results from problem (2d), find a threshold where you detect as many images with the new unwanted class but keep as many as possible of the images with the desired classes. Report the number of unwanted images you detect and the number of wanted images you remove in the process.
- (2f) Bonus problem: Pass the unlabeled part of the dataset through your trained model and predict known and unknown classes based on the threshold from problem (2e). Store the predictions as a binary vector in a Numpy array and upload it as part of the exam submission. After the exam, the predictions will be evaluated and the one who detects the most samples of the new class while keeping as much as possible of the known classes will receive a lot of glory and a little prize. The Numpy array can be saved using the `numpy.save` function, and please use the `.npz` extension. Also remember not to shuffle the data when doing your predictions.

Problem 3

In this problem, you will train an image classifier on the provided traffic sign dataset and explore several approaches to explain the model's predictions and detect backdoors. The data can be found on Canvas in `Canvas/Files/homeexam/problem3.zip`.

- (3a) Explain one approach that can be used to interpret/explain a specific prediction of a deep learning model and one that can be used to interpret/explain the model as a whole.
- (3b) Train a ResNet-18 to classify the data. In this problem, you are allowed to use existing frameworks like e.g. Torchvision, but without pretrained weights. Describe the network and your implementation and report the obtained accuracy for the training and validation data. For this, use the data contained in the "train" folder and split it into a 80%/20% train/validation split. Split the data per class so that you keep the same class distribution.
- (3c) Both quantitatively and qualitatively inspect the results on the provided test images (in the "test" folder) and compare these results to what you obtained on the validation set. Discuss what you observe.
- (3d) For the class where the model fails on the test-set, produce Class Saliency Maps for the training images as described in (3) (Section 3.1) to discover what the model bases its predictions on. Provide the saliency maps of 10 training images and comment on the observed results.
- (3e) For the same training images, perform also an Occlusion analysis as described in (4) (Section 4.2) where you occlude a 10x10 region and monitor the classifier output as you occlude different parts of the image. Provide illustrations of the same examples as in 3c) that mimic Figure 7 (e) in (4) and comment on the observed results.
- (3f) Based on your observations, propose an approach to modify the training data to remove the backdoor in the training dataset.

- (3g) Retrain your classifier on the modified data and report the new results for the test images. Discuss the results.
- (3h) Provide the saliency maps and the occlusion analysis maps for the new model for the same images as in 3d) and 3e). Discuss the results.

References

- [1] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. In *International Conference on Learning Representations*, 2015.
- [2] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *The Journal of Machine Learning Research*, 15(1):1929–1958, 2014.
- [3] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. Deep inside convolutional networks: Visualising image classification models and saliency maps. *arXiv preprint arXiv:1312.6034*, 2013.
- [4] Matthew D Zeiler and Rob Fergus. Visualizing and understanding convolutional networks. In *Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part I 13*, pages 818–833. Springer, 2014.